iDEA: an open-source Python framework for exact many-electron quantum mechanics and density functional theory in one dimension

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Abstract

We present iDEA (interacting Dynamic Electrons Approach), a free and open-source Python framework for the exact solution of the static and time-dependent many-electron Schrödinger equation for finite one-dimensional systems of a few electrons, together with approximate methods (non-interacting, Hartree, Hartree–Fock, a local density approximation and hybrid functionals) that share a common interface. Exact Kohn–Sham, and more generally exact fictitious-system, potentials are obtained by reverse engineering the exact ground-state or time-dependent density. iDEA is designed for reproducibility, interactivity and simplicity. It has been used to provide exact benchmarks for density functional and many-body perturbation theory, to generate training data for machine learning, and to teach many-body quantum physics. We describe the methods and their implementation, validate the exact solver against analytic results, and illustrate its use with three examples whose figures are regenerated by a script distributed with the code.

Keywords: exact diagonalisation, density functional theory, time-dependent DFT, exact Kohn–Sham potential, reverse engineering, one-dimensional model systems

Program summary

Program title: iDEA (interacting Dynamic Electrons Approach), version 1.1.0
CPC Library link to program files: (to be added by Technical Editor)
Developer's repository link: `https://github.com/JackWetherell/iDEA`
Licensing provisions: GPLv3 (or later)
Programming language: Python (≥ 3.8; continuous integration with 3.10, also run with 3.13)
Supplementary material: none
External routines/libraries: NumPy [1], SciPy [2], Matplotlib, tqdm, JupyterLab; optionally CuPy [3] for GPU acceleration.
Nature of problem: Exact ground, excited and time-dependent many-electron wavefunctions of finite one-dimensional systems with arbitrary external potentials, interactions and spin configurations are needed as reference data against which approximate methods, such as Kohn–Sham density functional theory (DFT), time-dependent DFT and Hartree–Fock theory, can be assessed. The exact Kohn–Sham potential corresponding to a given interacting density is also required.
Solution method: The many-electron Hamiltonian is represented on a uniform real-space grid using high-order finite differences, and the lowest eigenstates are found by sparse Lanczos diagonalisation followed by explicit antisymmetrisation of the space–spin wavefunction. Time propagation applies the matrix exponential of the Hamiltonian at each step. Approximate methods share a self-consistent single-particle solver. Exact Kohn–Sham potentials are found by iteratively inverting the ground-state or time-dependent density.
Additional comments including restrictions and unusual features: One spatial dimension, uniform grid with hard-wall boundaries. The cost and memory of the exact solver grow as N_x^N (and the antisymmetrised wavefunction as $(2N_x)^N$), restricting it in practice to two electrons on a few hundred grid points or three electrons on a few tens of points. The LDA is that of the spin-polarised electron gas. GPU acceleration applies to the exact solver only. An experimental Kohn–Sham strictly-correlated-electrons module is included. Installable via `pip install iDEA-latest`; documentation at `https://idea-interacting-dynamic-electrons-approach.readthedocs.io` and a tutorial notebook are provided with the source code.

1. Introduction

Approximate electronic-structure methods are routinely assessed against more accurate ones, but in three dimensions exact many-body solutions exist for only the smallest systems. Finite one-dimensional (1D) model systems offer an alternative: with a softened Coulomb interaction [4] they show many qualitative features of real atoms, molecules and materials [5, 6, 7], yet the many-electron Schrödinger equation can be solved exactly for a few electrons. Exact solutions give access to the quantities that approximate theories try to model, most notably the exact Kohn–Sham (KS) potential of density functional theory (DFT) [8, 9] and its time-dependent counterpart [10, 11, 12]. These are unknown in general and must be obtained by inverting the exact density [13, 14, 15, 16, 17, 18, 19, 20].

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ıDEA grew out of this programme in the group of R. W. Godby at the University of York. Exact time-dependent solutions and reverse-engineered KS potentials computed with ıDEA revealed static and dynamic steps in the exact exchange–correlation (xc) potential [21, 22, 23, 24], complementing related studies by others [25, 26]. The code has since been used to construct local density approximations (LDAs) from finite systems [27, 28]; to assess hybrid functionals [29], the GW approximation [30], the nearsightedness of many-body perturbation theory relative to KS DFT [31] and generalized KS theory [32]; to compute exact potentials for the fundamental gap [33]; and to generate exact one-body density matrices for machine learning [34], a use of exact 1D data pioneered elsewhere [35, 36]. It is also used in teaching.

The current version, ıDEA 1.1.0, is a rewrite of the original research code as a pip-installable Python package with a small, uniform interface designed for reproducibility, interactivity (for example in Jupyter notebooks) and simplicity. Octopus offers exact many-body solutions of model systems within a large real-space TDDFT code [37, 7], density-matrix renormalisation group calculations give 1D benchmarks for larger systems [5, 6], other packages treat lattice or second-quantised models [38, 39] or three-dimensional systems approximately [40], and n2v inverts densities supplied by other codes [18]. ıDEA combines exact real-space solutions, the standard approximate methods and density inversion, ground state and time dependent, in one lightweight package.

2. Methods and implementation

Systems. A `System` is defined by a uniform grid x_i with spacing δx, an external potential $v_{\mathrm{ext}}(x)$, an interaction matrix $v_{\mathrm{int}}(x, x')$ and a string of electron spins, e.g. `"uud"`. All examples here use the softened interaction $v_{\mathrm{int}}(x, x') = 1/(|x-x'|+1)$ used in earlier ıDEA work (e.g. Ref. [22]) (Hartree atomic units), a variant of the soft-Coulomb form $1/\sqrt{(x-x')^2+1}$ [4]; both are provided. The kinetic energy is represented by central finite differences of 3 to 13 points (default 13). The truncated stencil implies hard walls at the grid edges, so the box must be large enough for the density to vanish there.

Exact solution. For N electrons the spatial Hamiltonian

$$\hat{H} = \sum_{i=1}^{N} \hat{h}_i + \sum_{i<j} v_{\mathrm{int}}(x_i, x_j), \qquad \hat{h} = -\tfrac{1}{2}\tfrac{\mathrm{d}^2}{\mathrm{d}x^2} + v_{\mathrm{ext}}(x), \tag{1}$$

is assembled as a sparse matrix of dimension N_x^N: the one-body part is a Kronecker sum, $\sum_i I\otimes\cdots\otimes h\otimes\cdots\otimes I$, and the interaction is diagonal. The lowest eigenpairs are found with the implicitly restarted Lanczos method (ARPACK [41] via SciPy [2]). Each spatial eigenvector ψ is multiplied by the spin product state χ defined by the spin string, and the antisymmetriser $\sum_P (-1)^P \hat{P}$ is applied explicitly over all $N!$ permutations of the space–spin coordinates. Spatial states of the wrong exchange symmetry vanish and are discarded, so the kth surviving state is the kth fermionic eigenstate with the given S_z; the spin string fixes S_z, not the total spin, so for `"ud"` the excited states include the $S_z = 0$ triplets. The number of Lanczos eigenpairs needed is estimated heuristically from N, the spin string and k, and can be set explicitly.

Time propagation in the presence of a perturbation $\delta v(x, t)$ uses $\Psi(t_j) = \exp[-\mathrm{i}\,\delta t\,\hat{H}(t_j)]\,\Psi(t_{j-1})$, which is exact for a time-independent Hamiltonian and first order in δt otherwise. On the CPU the action of the exponential is evaluated with the algorithm of Al-Mohy and Higham [42]. Optionally, Hamiltonian construction, diagonalisation (CuPy's Lanczos solver) and propagation (a scaled Taylor series) run on a GPU through CuPy [3]. In single-run benchmarks distributed with the code, an NVIDIA RTX 4090 reduces the two-electron ground-state solve on a 450-point grid from about 260 s on an Intel i9-11900K CPU to about 17 s.

Approximate methods. Non-interacting electrons, Hartree, restricted and unrestricted Hartree–Fock (HF) [43], the LDA and a hybrid share one self-consistent single-particle solver. For each spin channel the Hamiltonian $h_\sigma[n, \rho_\sigma]$ is diagonalised densely, and the density n and spin-resolved density matrices $\rho_\sigma(x, x')$ are mixed linearly (default mixing 0.5) until $\int |n - n_{\mathrm{old}}|\,\mathrm{d}x$ falls below a tolerance (default 10^{-10}). There is no iteration limit, and systems prone to charge sloshing need smaller mixing; the examples below use 0.2. The restricted option is intended for closed shells. Each method supplies its potential, namely v_{H}, $v_{\mathrm{H}} + v_{\mathrm{x}}^{\mathrm{HF}}$, $v_{\mathrm{H}} + v_{\mathrm{xc}}^{\mathrm{LDA}}$ or $v_{\mathrm{H}} + \alpha v_{\mathrm{x}}^{\mathrm{HF}} + (1-\alpha) v_{\mathrm{xc}}^{\mathrm{LDA}}$ (default $\alpha = 0.8$), and a total-energy expression. The LDA is the exchange–correlation energy of the spin-polarised 1D homogeneous electron gas parametrised in Refs. [27, 28], evaluated on the total density, so it is appropriate for fully spin-polarised systems. Excited configurations are formed by re-occupying the self-consistent orbitals, ranked by the sum of orbital energies; they are not separately self-consistent. Time-dependent versions (TDHF, adiabatic LDA, etc.) propagate the occupied orbitals with the same exponential, rebuilding the Hamiltonian from the density and density matrix of the previous step. An experimental implementation of KS strictly-correlated electrons [44, 45, 46, 47], contributed by D. Kooi, uses the same solver but needs the interaction and its derivatives as functions, is not yet part of the common interface, and can fail for densities with long low-density tails.

Reverse engineering. For a target density $n_{\mathrm{T}}(x)$ and a chosen method, ıDEA finds the external potential of the fictitious system that reproduces n_{T} by iterating

$$v^{(i+1)}(x) = v^{(i)}(x) + \mu\left[n^{(i)}(x)^p - n_{\mathrm{T}}(x)^p\right] \tag{2}$$

until $\int |n^{(i)} - n_{\mathrm{T}}|\,\mathrm{d}x$ is below a tolerance (defaults $\mu = 1$, $p = 0.1$), a scheme used in earlier ıDEA work [22, 23] and akin to that of van Leeuwen and Baerends [13]. The power $p < 1$ enlarges corrections where the density is small. With the non-interacting method and an exact n_{T} this yields the exact KS potential v_{KS}, from which $v_{\mathrm{xc}} = v_{\mathrm{KS}} - v_{\mathrm{ext}} - v_{\mathrm{H}}$; the arbitrary constant is fixed afterwards, for example by requiring the highest occupied KS eigenvalue to equal minus the ionisation

Table 1: Validation of the exact solver and indicative cost. (a) Two electrons in a harmonic well ($\omega = 0.3275$, $N_x = 231$, $L = 20$) compared with the separated centre-of-mass/relative-coordinate reference on the same grid: absolute energy error, maximum density error and wavefunction infidelity $1 - |\langle\Psi_{\mathrm{ref}}|\Psi\rangle|$. (b) Kohn-mode excitation energy (two uu electrons, $\omega = 0.25$, $N_x = 151$, $L = 20$); density error of a single electron oscillating in a harmonic well over 500 steps; and the centre-of-mass error of two uu electrons ($\omega = 0.25$, $N_x = 101$, $L = 20$) driven by $\delta v = 0.02\sin(0.4t)\,x$ for $0 \le t \le 30$, against the harmonic-potential theorem. (c) Grid convergence of the ground-state energy of the atom in Fig. 1. (d) Wall time of `interacting.solve` on 4 cores of an Intel Xeon (2.8 GHz).

(a) Two-electron ground state

	singlet (ud)	triplet (uu)
E (Ha)	0.779893	0.936028
$\lvert\Delta E\rvert$	3.7×10^{-12}	1.6×10^{-12}
$\max\lvert\Delta n\rvert$	1.1×10^{-11}	4.0×10^{-12}
infidelity	5.8×10^{-15}	3.7×10^{-14}

(b) Excitations and dynamics

$E_1 - E_0$ vs $\omega = 0.25$	0.25000004	error 4.0×10^{-8}
TD density, $\max\lvert\Delta n(x,t)\rvert$	2.0×10^{-13}	
driven HPT, $\max\lvert\Delta x_{\mathrm{cm}}\rvert$, $\delta t = 0.05$	4.0×10^{-3}	(amplitude 0.53)
$\delta t = 0.025$	2.0×10^{-3}	

(c) Grid convergence, atom (uu)

$\delta x = 0.200$	$E = -1.34519$
$\delta x = 0.133$	$E = -1.34131$
$\delta x = 0.100$	$E = -1.33995$

(d) Ground-state solve time

$N = 2$: N_x	60, 100, 140, 180, 220
time (s)	0.2, 1.4, 3.4, 9.7, 21.5
$N = 3$: N_x	20, 30, 40
time (s)	1.1, 6.9, 19.7

energy. In the time-dependent case, $\delta v(x, t_j)$ is found at each step by solving, with SciPy's Powell hybrid method, for the potential for which one propagation step reproduces $n_{\mathrm{T}}(x, t_j)$. The root finder's convergence is not enforced, so the returned per-step density error should be checked. This density-based scheme is practical on small grids; earlier iDEA studies used a current-density-based variant [22].

Observables and interface. The `observables` module computes densities, density matrices, and Hartree and exchange potentials and energies from many-body states, single-particle states or their time evolution. Each method module provides `solve`, `propagate`, `total_energy` and `hamiltonian`, so the single-particle methods can be looped over (their `solve` signatures coincide; the exact solver has its own options):

```
import numpy as np, iDEA
x = np.linspace(-12, 12, 181)
s = iDEA.system.System(x, -2/(abs(x)+1),
        iDEA.interactions.softened_interaction(x),
        electrons="uu")
gs = iDEA.methods.interacting.solve(s, k=0)
n = iDEA.observables.density(s, state=gs)
for m in iDEA.iterate_sb_methods:   # NI, H, HF, LDA, hybrid
    st = m.solve(s, mixing=0.2)
    print(m.name, m.total_energy(s, st))
ks = iDEA.reverse_engineering.reverse(
        s, n, iDEA.methods.non_interacting, tol=1e-8)
v_xc = ks.v_ext - s.v_ext \
       - iDEA.observables.hartree_potential(s, n)
```

3. Validation

The exact solver is tested against a semi-analytic reference (Table 1a). For two electrons in a harmonic potential $v_{\mathrm{ext}} = \frac{1}{2}\omega^2 x^2$ with the softened interaction, the Hamiltonian separates into an analytic centre-of-mass part and a relative-coordinate problem, which the reference solves with the same stencil and grid spacing. Agreement to $\sim 10^{-12}$ for both the spin-singlet (ud) and spin-triplet (uu) ground states therefore validates the many-body Hamiltonian, eigensolver and antisymmetrisation independently of discretisation error. Discretisation error is controlled by the grid spacing: for the atom of Fig. 1, whose potential has a cusp at the origin, the energy changes by about 10^{-3} Ha between $\delta x = 0.133$ and 0.100 (Table 1c). The harmonic-potential theorem [48] gives two further tests. Interacting electrons in a harmonic well have a centre-of-mass (Kohn) excitation of energy exactly ω, which iDEA reproduces for the uu case ($N_x = 151$, $L = 20$) to within the small discretisation error that breaks exact separability. Under a uniform time-dependent field the interacting density moves rigidly, with a centre of mass that obeys the classical driven-oscillator equation; the exact propagation of two uu electrons reproduces this trajectory, with an error that halves when δt is halved, as expected for the first-order treatment of a continuously varying field (Table 1b). Finally, a single electron displaced in a harmonic well oscillates with an analytic density that the propagator reproduces at every step. The two-electron ground-state tests run in continuous integration; the remaining tests, and tests of the self-consistent methods (energy identities, agreement of restricted and unrestricted HF for a closed shell, and a reverse-engineering round trip), are run with `pytest`. The approximate methods are not compared with independent reference implementations.

4. Examples

All results below are produced by `paper/scripts/make_figures.py` in the repository, which regenerates every figure and number in this paper; timings in this section and Table 1d are for 4 cores of an Intel Xeon (2.8 GHz) cloud machine, not the machine used for the GPU benchmarks above.

Ground state: comparing approximations. Figure 1 shows two spin-polarised electrons bound by a softened nucleus of charge 2. The exact ground-state energy is $E = -1.3413$ Ha. HF (-1.3388 Ha), the LDA (-1.3348 Ha) and the hybrid (-1.3379 Ha) are close to it, whereas Hartree theory (-0.8852 Ha) and non-interacting electrons (-1.6388 Ha) are not. The integrated density errors $\int |n - n_{\mathrm{exact}}|\,\mathrm{d}x$ are 0.005 (HF), 0.009 (hybrid), 0.050 (LDA), 0.418 (non-interacting) and 0.654 (Hartree). The self-interaction in the Hartree potential makes its density far too diffuse, worse even than ignoring the interaction. The LDA removes most of this error but still moves density from $|x| \approx 2$ to $|x| \approx 4$ [49], while HF, which is free of one-electron self-interaction, gives the most accurate density.

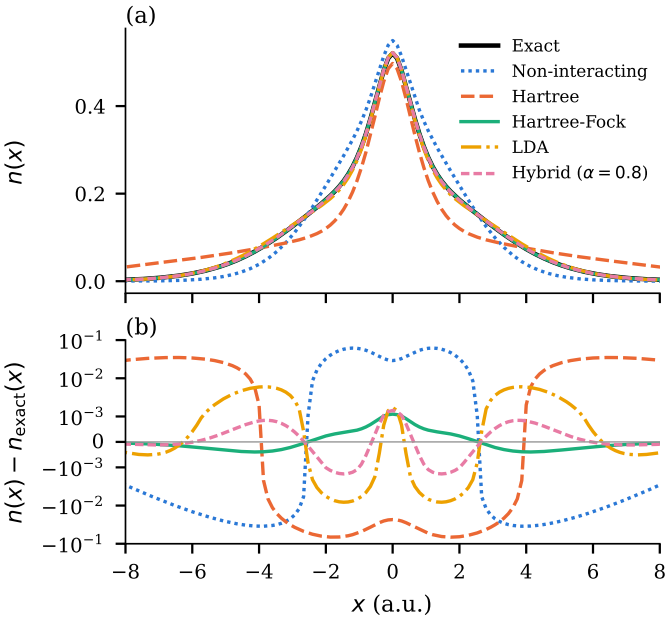


Figure 1: (a) Ground-state density of two spin-polarised electrons in the softened atom $v_{\text{ext}} = -2/(|x| + 1)$ ($N_x = 181$, $\delta x = 0.133$), exact and from each approximate method. (b) Error relative to the exact density (linear within $\pm 10^{-3}$, logarithmic beyond).

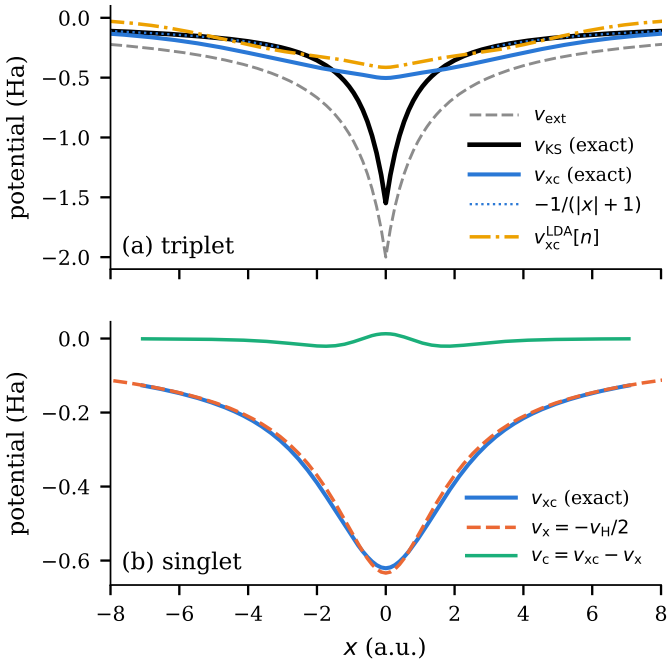


Figure 2: Exact KS and xc potentials obtained by reverse engineering exact densities of the softened atom of Fig. 1, with the constant fixed so that the highest occupied KS eigenvalue equals minus the exact ionisation energy, and shown only where $n > 10^{-5}$. (a) Spin-polarised (uu) electrons, compared with the LDA xc potential of the same density. (b) Spin-singlet (ud) electrons, for which the exact exchange potential is $-v_{\text{H}}/2$, giving the exact correlation potential v_{c}.

Exact Kohn–Sham potentials. Reverse engineering the exact density of the spin-polarised atom (Fig. 2a) gives the exact KS potential, and hence v_{xc}; the KS density reproduces the exact one to $\int |n_{\text{KS}} - n| \, \mathrm{d}x = 1.0 \times 10^{-8}$. With the constant fixed by the exact ionisation energy (0.202 Ha), the exact v_{xc} tends towards the expected $-1/(|x| + 1)$ away from the atom (-0.133 against -0.111 Ha at $x = -8$, where $n \approx 3 \times 10^{-3}$); the residual difference reflects the slow approach to the asymptotic region and the finite box. The LDA, which depends only on the local density, decays far too quickly (-0.029 Ha) and is also too shallow at the centre (-0.414 against -0.503 Ha). For the spin-singlet atom (Fig. 2b; $E = -1.7162$ Ha) the exact exchange potential of two electrons in one spatial orbital is $-v_{\text{H}}/2$, so reverse engineering isolates the exact correlation potential, which lies between -0.021 and 0.013 Ha where $n > 10^{-3}$. The same procedure, with another method in place of the non-interacting solver, gives exact Hartree- or HF-like fictitious systems.

Dynamics. Figure 3 shows the free dipole oscillation of the same atom after a short field pulse. The exact propagation of 1000 steps takes about 67 s. Because the field is piecewise constant, the exact propagation is independent of δt here (halving it changes $d(t)$ only at round-off level); the approximate methods, which update their Hamiltonian explicitly, are first order in δt. The approximate methods, propagated through the same interface from their own ground states, differ in both frequency and amplitude: the adiabatic LDA oscillates too fast and time-dependent Hartree theory too slowly, while TDHF and the hybrid follow the exact dipole closely. The exact time-dependent density can in turn be reverse engineered to give the exact time-dependent KS potential [22, 23].

5. Conclusions

iDEA provides exact many-electron solutions of 1D model systems, the standard approximate methods and exact reverse engineering of KS and other fictitious-system potentials, ground state and time dependent, behind a small Python interface. Because the wavefunction is stored on an N_x^N grid, the exact solver is limited in practice to two electrons on a few hundred points or three electrons on a few tens of points (Table 1d). This suffices for the reference calculations that motivate the code and lets them run interactively. The simple update of Eq. (2) converges for the systems shown here, but its step size may need reducing, and it can stall for weakly coupled subsystems whose KS orbitals are nearly degenerate. More robust inversion algorithms [15, 16, 17, 18], convergence safeguards in the self-consistent and inversion loops, the many-body kinetic energy (currently unavailable for exact states), spin-resolved functionals and symmetry-adapted or tensor-network representations for more electrons are natural extensions. iDEA is developed openly, and contributions are welcome.

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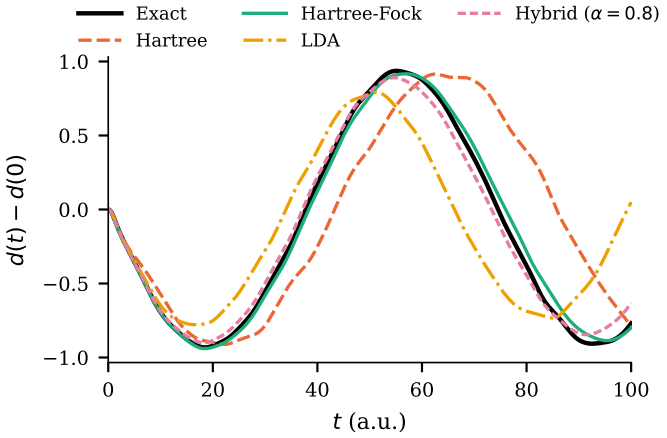


Figure 3: Dipole moment $d(t) = \int x\, n(x,t)\, \mathrm{d}x$ of the atom of Fig. 1 after a uniform-field pulse of strength 0.05 a.u. and duration 1 a.u., from exact propagation and from the time-dependent Hartree, HF, adiabatic LDA and hybrid methods ($N_x = 121$, $\delta t = 0.1$).

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